

# REFERRING MULTI-OBJECT TRACKING IN MOVING-CAMERA VIDEOS VIA GLOBAL MOTION COMPENSATION

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## ABSTRACT

Referring multi-object tracking (RMOT) takes a video and a natural-language expression as input and tracks all objects described by the expression. Many expressions describe how an object moves rather than how it appears. In a video captured by moving cameras, a parked vehicle may appear to move, while a vehicle moving with the camera may show little displacement. Existing motion-language RMOT methods align motion with text but do not explicitly remove camera-induced motion. In this paper, we propose extracting residual motion across frames and comparing motion features with the query language expression. We consider the motion-matching extent and integrate it with the RMOT method’s prediction result. In the evaluation, we verify the performance gain by taking the proposed motion compensation module as a plug-in.

**Index Terms**— referring multi-object tracking, ego-motion compensation, motion-language alignment, plug-in module, Refer-KITTI, video grounding

## 1. INTRODUCTION

Referring multi-object tracking (RMOT) outputs tracking results of the objects described by a free-form language expression. For example, a “left-vehicles-in-red” description targets the red vehicles on the left. However, many descriptions focus on object motion, such as “moving cars” or “a vehicle turning left”. In videos captured by a moving camera, the observed motion is a mixture of object motion and camera motion, and thus a static car may appear to move, while a moving car may appear static across video frames.

Previous RMOT methods [5, 1, 3] match language expressions to objects. VMRMOT [6] considers explicit motion cues, such as changes in direction and speed, in the matching process. All of these methods, however, compute motion from the object’s bounding-box displacement across frames, which mixes object motion with camera motion in a video captured by moving cameras.

Conventional multi-object tracking (MOT) methods already proposed to remove camera motion: BoT-SORT [7] estimates camera-induced image motion, while UCMCTrack [8] stabilizes object trajectories. How motion compensation can be extended to RMOT is still underexplored.

We propose separating camera motion from object motion and providing motion information for language-guided tracking. It consists of four stages. First, a homography, which describes the geometric transformation across frames, is estimated to measure camera motion. The estimated homographies are accumulated across frames to reliably measure object motion. Second, residual velocity, which represents object motion after removing camera motion, is computed across frames and used as a temporal motion feature. Third, a Contrastive Motion Aligner is developed to compare motion features with language expressions and produce a motion-matching score. Finally, the motion-matching score is rescaled to the host model’s score range and added to the host’s matching score. Here, the host model refers to a two-stage RMOT method we use as a baseline, such as iKUN [1] and FlexHook [3]. The proposed process is taken as a plug-in to provide an extra motion-matching score. We thus improve existing RMOT baselines without retraining them.

The accumulated homographies are the key component. By compensating for camera motion, they allow the motion feature to better represent the object’s actual movement rather than the apparent motion induced by camera movement. They also preserve motion information across frames, enabling the model to capture motion patterns described by expressions such as “moving” or “turning”.

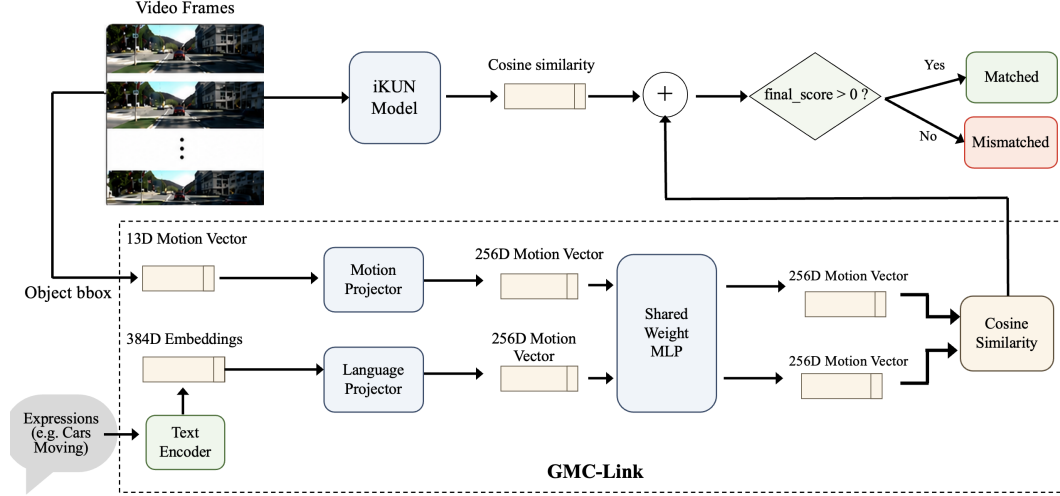
This work connects ego-motion compensation with motion-language modeling for RMOT. Our main contributions are as follows.

- We propose a plug-in that aligns ego-compensated residual velocity with the referring expression, thereby improving the host RMOT method without modifying or retraining it.
- Based on iKUN, the proposed plug-in raises moving-class HOTA by 8.83. The performance gains can be observed in various settings.

## 2. RELATED WORKS

### 2.1. Referring MOT and motion-language fusion

TransRMOT [5] introduced the RMOT task on the Refer-KITTI dataset. Later methods improved different parts of



**Fig. 1.** The host model, iKUN in this case, provides tracking scores. The GMC-Link estimates camera motion, derives residual motion, aligns it with language, and provides an extra cue to adjust the tracking scores.

this task. iKUN [1] separated language-based object matching from object tracking, while FlexHook [3] improved how visual and language information interact. ReferGPT [10] used a multimodal large language model to describe existing object tracks, and MEX [11] provided a memory-efficient module for matching tracks with language.

More recent RMOT methods [12, 13, 14, 15] further improve models’ understanding of object descriptions. VMR-MOT [6] explicitly represents motion using information such as position, direction, changes in distance, and changes in speed. However, these methods still estimate object motion from the object’s bounding-box movements across frames. When the camera moves, the observed bounding-box movement contains both object motion and camera motion.

## 2.2. Camera and ego-motion compensation in tracking

Camera motion compensation is widely used in traditional multi-object tracking (MOT). BoT-SORT [7] estimates camera-induced motion across frames. UCMCTrack [8] estimates how the camera moves relative to the road surface. EMAP [18] further separates the camera’s turning and movement from object motion within the Kalman prediction process.

These methods demonstrate that compensating for camera motion can improve tracking objects across frames. However, the compensated positions are mainly used to match objects between adjacent frames. They are not accumulated over time to describe an object’s motion, nor are they connected to language expressions. Our method builds on established techniques for separating camera motion from object motion, including plane-plus-parallax analysis [19], homography estimation [20], motion segmentation under moving cameras [21], and ego-motion estimation in visual odometry [22]. We do not propose a new camera-motion estimation technique. In-

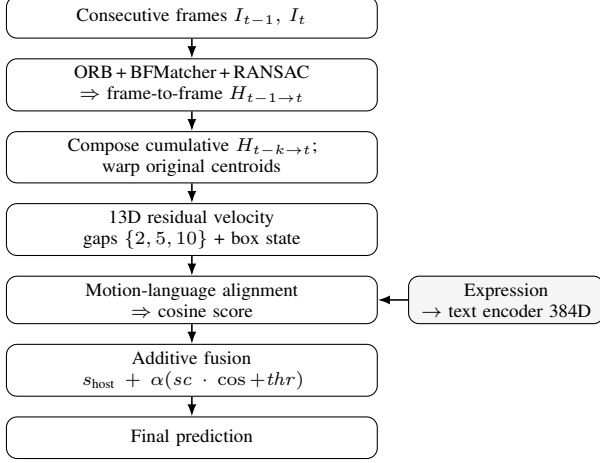
stead, we convert camera-compensated trajectories into motion features and align them with language for RMOT.

## 3. METHOD

Figure 1 shows the proposed RMOT framework with the global motion compensation (GMC) module. The GMC module is attached to a two-stage RMOT method, iKUN [1] in this case. iKUN performs object detection and tracking and then outputs appearance-language matching scores. The inputs to the GMC module are the sequence of object bounding boxes across frames, the video frames, and the language expression. Based on these inputs, the GMC module estimates camera motion, extracts residual object motion, aligns it with the language expression, and produces a matching score. The score is then combined with the host model’s predicted score, as an embodiment of the late fusion scheme. The complete pipeline of the GMC module is illustrated in Figure 2, which consists of four stages: ego-motion compensation, multi-scale residual velocity extraction, motion-language alignment, and late fusion.

### 3.1. Ego-motion compensation

This stage estimates camera motion from consecutive video frames so that object motion can be separated from camera-induced motion. We first extract ORB [16] keypoints from video frames. Keypoints inside objects’ bounding boxes are removed so that moving foreground objects do not affect camera-motion estimation. The remaining keypoints at the  $(t - 1)$ -th frame  $I_{t-1}$  and the  $t$ -th frame  $I_t$  are matched, and the RANSAC [17] algorithm is adopted to estimate the homography matrix  $H_{t-1 \rightarrow t}$  between them, which describes the geo-



**Fig. 2. GMC-Link pipeline.** ORB homographies compose into a cumulative transform, produce a 13D motion feature, align with the expression, and correct the host score through additive fusion.

metric transformation caused by camera motion from frame  $I_{t-1}$  to frame  $I_t$ .

To describe camera motion across frames, the estimated homographies are accumulated as

$$H_{t-k \rightarrow t} = H_{t-1 \rightarrow t} \times H_{t-2 \rightarrow t-1} \times \cdots \times H_{t-k \rightarrow t-k+1}, \quad (1)$$

where  $H_{t-k \rightarrow t}$  represents the accumulated camera transformation from frame  $I_{t-k}$  to frame  $I_t$ . Instead of repeatedly transforming already warped coordinates, the accumulated homography is applied only once to the original object centroid. This reduces numerical drift and provides a stable reference for estimating ego velocity in the next stage.

### 3.2. Multi-scale residual velocity

Using the accumulated homography matrix and the objects' bounding boxes, this stage estimates the residual velocity, which represents object motion after removing the estimated camera motion. For a temporal gap  $g$ , the raw velocity  $v_g^{\text{raw}}$  is derived from the displacement of the object centroid between two frames ( $I_{t-g}$  and  $I_t$ ) and is therefore mixed with object motion and camera motion. To remove the camera motion, we apply the accumulated homography matrix to an object's centroid  $\mathbf{o}_{t-g}$  at  $I_{t-g}$  to estimate its centroid  $\hat{\mathbf{o}}_t$  at  $I_t$ . The distance between the estimated centroid  $\hat{\mathbf{o}}_t$  and the real centroid  $\mathbf{o}_t$  is calculated and is divided by the time difference  $g$  to get the ego velocity  $v_g^{\text{ego}}$ . The residual velocity  $v_g^{\text{res}}$  is then computed as

$$v_g^{\text{res}} = v_g^{\text{raw}} - v_g^{\text{ego}}, \quad (2)$$

This representation approximates the motion that a stationary camera would observe. For example, a parked vehicle has near-zero residual velocity even when camera motion causes

a large image displacement. All velocities are normalized by the image dimensions to make them independent of image resolution.

To capture motion over different time spans, residual velocities are computed at temporal gaps of  $\{2, 5, 10\}$  frames. That means the accumulated homography matrices mentioned in Eqn. (1) have three versions, i.e.,  $H_{t-2 \rightarrow t}$ ,  $H_{t-5 \rightarrow t}$ , and  $H_{t-10 \rightarrow t}$ . The residual velocities are concatenated with the object's box information to form a 13-dimensional motion feature,

$$((dx_s, dy_s), (dx_m, dy_m), (dx_l, dy_l), \Delta w, \Delta h, c_x, c_y, w, h, \rho), \quad (3)$$

where the first six values are the residual velocities at the three temporal gaps, and  $\Delta w, \Delta h$  capture the change in box scale. The terms  $c_x$  and  $c_y$  describe the coordinates of the box center, and  $w$  and  $h$  describe the width and height of the box, respectively. The final term  $\rho$  is the ratio of the object's residual speed at the middle gap to the magnitude of the estimated camera motion.

### 3.3. Motion-language alignment

Given the motion features mentioned above and the language expression, this stage measures how well the observed object motion matches the described motion. The 13-dimensional motion feature is encoded by a motion encoder, while the language expression is encoded by all-MiniLM-L6-v2 (MiniLM) [23] into the text embeddings. Both embeddings are projected into the same feature space using linear projection layers and a shared Multi-Layer Perceptron (MLP).

The model is trained using Information Noise Contrastive Estimation (InfoNCE) [24] loss to maximize the similarity between matched motion-expression pairs while separating unmatched pairs. A positive pair is formed by the embeddings of an object and the corresponding expression. Other object-expression pairs in the same mini-batch are treated as negatives. The MiniLM text encoder is kept fixed, and only the projection layers and motion encoder are trained. During inference, the cosine similarity  $s_{\text{gmc}}$  between an object's motion embeddings and the target expression's embeddings is used as the matching score and passed to the late fusion stage described below.

### 3.4. Fusion

The matching score is combined with the prediction score of the host RMOT model to obtain the final prediction:

$$s_{\text{final}} = s_{\text{host}} + \alpha(s_{\text{gmc}} + thr), \quad (4)$$

where  $s_{\text{host}}$  is the prediction score produced by the host model,  $thr$  adjusts the decision threshold, and  $\alpha$  controls the contribution of the GMC module.

Since the GMC module only adds extra influence on the final prediction score, it can be integrated into existing two-

**Table 1.** Performance in terms of the pooled HOTA on the Refer-KITTI test dataset.

| Host            | Baseline | w/ GMC module      | Gain             |
|-----------------|----------|--------------------|------------------|
| iKUN [1]        | 44.564   | $44.634 \pm 0.066$ | $+0.070$         |
| FlexHook V1 [3] | 53.824   | $53.526 \pm 0.087$ | $-0.298^\dagger$ |
| FlexHook V2 [3] | 42.526   | $42.807 \pm 0.038$ | $+0.281$         |

stage RMOT frameworks without changing or retraining their detector, tracker, visual encoder, or language encoder.

## 4. EXPERIMENTS

### 4.1. Setup

We evaluate the proposed module on the Refer-KITTI V1 [5] and the Refer-KITTI V2 [4] benchmarks using the HOTA metric from TrackEval [2], which jointly measures detection and association performance. We report both moving-class HOTA, which evaluates only expressions referring to moving objects, and pooled HOTA, which evaluates all expressions in the benchmark.

We attach the proposed GMC module to iKUN [1] and FlexHook [3]. We also compare against a raw-velocity variant that directly uses image-plane motion without ego-motion compensation. The GMC module is trained using AdamW with a learning rate of  $10^{-3}$ , batch size 256, and 100 epochs. Camera motion is estimated using ORB with 1,500 keypoints and RANSAC with a 5-pixel threshold.

Refer-KITTI is highly imbalanced, with only about one-sixth of the language expressions referring to moving objects. Consequently, improvements in motion understanding may have only a limited impact on pooled HOTA. Therefore, moving-class HOTA is used as the primary evaluation metric, while pooled HOTA is reported to reflect overall benchmark performance.

### 4.2. Main results

Table 1 summarizes the pooled HOTA results on the Refer-KITTI dataset. We evaluate the performance variations when iKUN or FlexHook is used as the host model, with or without the GMC module. Table 1 shows that, when the GMC module is integrated, the performance of iKUN and FlexHook V2 is raised. These results show that compensated motion information can be beneficial to RMOT. Not surprisingly, we observe that the gains mainly come from motion-referring expressions. Therefore, we especially evaluate the influence of the GMC module on movement-related expressions in the following.

Table 2 shows performance variations specific to 27 movement-related expressions in the Refer-KITTI test dataset. iKUN, which has a lower baseline HOTA, achieves greater improvement ( $+8.83 \pm 0.83$ ). On the other hand, FlexHook V1 and V2 already have stronger baseline performances and

**Table 2.** Performance variations specific to movement-related expressions.

| Host            | Baseline | Performance gain |
|-----------------|----------|------------------|
| iKUN [1]        | 20.25    | $+8.83 \pm 0.83$ |
| FlexHook V1 [3] | 43.98    | $+0.91 \pm 0.79$ |
| FlexHook V2 [3] | 48.02    | $+0.43 \pm 0.17$ |

**Table 3.** Per-class HOTA ablation on iKUN.

| Variant                  | MOVING HOTA                        | STATIC HOTA                        |
|--------------------------|------------------------------------|------------------------------------|
| Native host (no module)  | 20.25                              | —                                  |
| Full module              | <b><math>29.08 \pm 0.83</math></b> | <b><math>43.26 \pm 0.26</math></b> |
| —ego (raw velocity)      | $27.31 \pm 0.66$                   | $41.19 \pm 0.39$                   |
| —multiscale (single gap) | $28.82 \pm 0.84$                   | $43.42 \pm 0.10$                   |
| — $\rho$ (zeroed slot)   | $29.74 \pm 0.59$                   | $43.35 \pm 0.35$                   |

smaller gains can be obtained ( $+0.91 \pm 0.79$  and  $+0.43 \pm 0.17$ , respectively).

### 4.3. Ablation Studies

Table 3 evaluates the contribution of each component in GMC-Link. The full module improves moving-class HOTA from 20.25 to 29.08 ( $+8.83$ ). When ego-motion compensation is removed, the score decreases to 27.31, indicating that simply using image-plane motion already provides useful motion cues, but camera motion compensation is still necessary for the best performance. Compared with this raw-velocity variant, the full module gains 1.77 moving-class HOTA with Welch’s t-test significance ( $p=0.006$ ).

Removing ego-motion compensation also reduces static-class HOTA from 43.26 to 41.19 ( $p<10^{-4}$ ). This is because stationary objects appear to move in the image when the camera moves, causing them to be incorrectly treated as moving objects. In contrast, removing the multi-scale gaps or the residual-to-background ratio  $\rho$  produces only small changes that stay within seed variation, suggesting that these components have a relatively limited impact compared with ego-motion compensation.

Figure 3 illustrates this behavior using a real example. The parked car moves across the image because of camera motion, while the moving car remains at nearly the same image location. Without ego-motion compensation, these two cases are easily confused because they have similar image-plane motion. After compensating for camera motion, the parked car has nearly zero residual motion, while the moving car retains nonzero residual motion. This allows GMC-Link to correctly distinguish between stationary and moving objects.

Finally, the complete module runs at 68 FPS (median 14.8 ms/frame, excluding host inference), indicating that the proposed module can be integrated into existing RMOT frameworks with low computational overhead.

## 5. CONCLUSION AND LIMITATIONS

GMC-Link is a decision-level plug-in module that improves motion grounding in referring multi-object tracking without modifying the detector, tracker, or appearance-language matching model. By compensating for camera motion and aligning residual motion with language descriptions, the proposed module helps distinguish object motion from camera-induced motion. Experimental results on Refer-KITTI show that GMC-Link improves moving-class HOTA in the evaluated host-split settings while maintaining competitive pooled HOTA, suggesting that it can enhance motion understanding without sacrificing overall tracking performance.

Despite these promising results, several limitations remain. First, the proposed camera motion compensation relies on ORB and RANSAC, which estimate a single homography and may become less accurate in scenes with strong parallax or large non-planar motion. Second, the calibration parameters are host-specific and need to be estimated for different host models. Finally, our experiments are conducted on two RMOT frameworks, and additional evaluations on more host architectures and datasets are needed to further validate the generality of the proposed approach.

In future work, we plan to investigate more robust camera motion estimation methods and richer motion features that can better handle complex dynamic scenes. We also hope to evaluate GMC-Link on a broader range of referring multi-object tracking frameworks to further improve its generalization.

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**Fig. 3. Real recovery under camera ego-motion** on “moving cars” (Refer-KITTI seq 0005, frames 85/94/100 of a forward-driving clip; the green car moves, the red car is parked). The dashed line marks the mover’s image column. **Red:** the parked car’s centroid slides left as the camera passes it (center- $x$  261  $\rightarrow$  104 px), so the camera-naïve host scores it “moving” (false positive), yet its ego-compensated residual is near zero and the module restores it to static. **Green:** the moving car holds its column (center- $x \approx 605$ ,  $< 1$  px/frame), so the host misses it (false negative), but its nonzero residual — unlike the static background — recovers it as moving. Both errors flip from the same host logit through the additive correction.

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